EXISTENCE AND PROPERTIES OF THE DUALIZING COMPLEX OF A PROPER SCHEME OVER A FIELD

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0FVV (sources/stacks-duality/pinned-original-source.tex line 6934).

Allowed standard prerequisites (editorial).

- The right adjoint of pushforward and the relative dualizing complex. Same chapter, Lemma labelled lemma-twisted-inverse-image and Remark labelled remark-relative-dualizing-complex, cited at proof lines 6965-6972. Defines omega and identifies it with f^! of the structure sheaf.
- Identification of the right adjoint with a dualizing complex. Same chapter, Lemma labelled lemma-shriek-dualizing, cited at proof lines 6974-6976. Makes omega a dualizing complex.
- Vanishing and support statements for a good dualizing complex, and its normalisation at a point. Same chapter, Lemmas labelled lemma-vanishing-good-dualizing and lemma-good-dualizing-normalized, cited at proof lines 6984-6986 and 6990-6992, and Dualizing Complexes, Lemma labelled dualizing-lemma-depth-in-terms-dualizing-complex, cited at proof lines 6992-6994. Give properties (1) to (4).
- The dualizing module of a proper scheme and the smooth case. Same chapter, Lemma labelled lemma-dualizing-module-proper-over-A, cited at proof lines 7000-7002, and Lemma labelled lemma-smooth-proper, cited at proof lines 7006-7008. Give properties (6) and (7).

Source reference. Original lemma 0A9E: sources/stacks-duality/pinned-original-source.tex, source lines 653–653. Same-chapter pointer into the author environment lemma at source line 653; invoked at proof line 6968 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Date: stacks-project:duality.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 0B6S: sources/stacks-duality/pinned-original-source.tex, source lines 2945–2945. Same-chapter pointer into the author environment remark at source line 2945; invoked at proof line 6973 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original section 0A9Y: sources/stacks-duality/pinned-original-source.tex, source lines 3940–3940. Same-chapter pointer into the author environment lemma at source line 3940; invoked at proof line 6976 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0AA3: sources/stacks-duality/pinned-original-source.tex, source lines 4708–4708. Same-chapter pointer into the author environment lemma at source line 4708; invoked at proof line 6977 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original section 0AWI: sources/stacks-duality/pinned-original-source.tex, source lines 6070–6070. Same-chapter pointer into the author environment example at source line 6070; invoked at proof line 6980 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original section 0AWJ: sources/stacks-duality/pinned-original-source.tex, source lines 6085–6085. Same-chapter pointer into the author environment example at source line 6085; invoked at proof line 6981 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0AWN: sources/stacks-duality/pinned-original-source.tex, source lines 6125–6125. Same-chapter pointer into the author environment lemma at source line 6125; invoked at proof line 6985 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0AWL: sources/stacks-duality/pinned-original-source.tex, source lines 5805–5805. Same-chapter pointer into the author environment lemma at source line 5805; invoked at proof line 6990 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0AWR: sources/stacks-chow/tags.txt, source lines 13729–13729. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 6992 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0AWP: sources/stacks-duality/pinned-original-source.tex, source lines 6157–6157. Same-chapter pointer into the author environment lemma at source line 6157; invoked at proof line 7002 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0BRT: sources/stacks-duality/pinned-original-source.tex, source lines 3879–3879. Same-chapter pointer into the author environment lemma at source line 3879; invoked at proof line 7007 of the counted proof.

It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let X be a proper scheme over a field k. There exists a dualizing complex $\omega_X^\bullet$ with the following properties*

(1) *$H^i(\omega_X^\bullet)$ is nonzero only for $i \in [-\dim(X), 0]$,*
(2) *$\omega_X = H^{-\dim(X)}(\omega_X^\bullet)$ is a coherent (S_2)-module whose support is the irreducible components of dimension $\dim(X)$,*
(3) *the dimension of the support of $H^i(\omega_X^\bullet)$ is at most $-i$,*
(4) *for $x \in X$ closed the module $H^i(\omega_{X,x}^\bullet) \oplus \ldots \oplus H^0(\omega_{X,x}^\bullet)$ is nonzero if and only if $\text{depth}(\mathcal{O}_{X,x}) \leq -i$,*
(5) *for $K \in D_{QCoh}(\mathcal{O}_X)$ there are functorial isomorphisms*[1]
$$\text{Ext}^i_X(K, \omega_X^\bullet) = \text{Hom}_k(H^{-i}(X, K), k)$$
compatible with shifts and distinguished triangles,
(6) *there are functorial isomorphisms $\text{Hom}(\mathcal{F}, \omega_X) = \text{Hom}_k(H^{\dim(X)}(X, \mathcal{F}), k)$ for $\mathcal{F}$ quasi-coherent on X, and*
(7) *if $X \to \text{Spec}(k)$ is smooth of relative dimension d, then $\omega_X^\bullet \cong \wedge^d \Omega_{X/k}[d]$ and $\omega_X \cong \wedge^d \Omega_{X/k}$.*

Proof. Denote $f : X \to \text{Spec}(k)$ the structure morphism. Let a be the right adjoint of pushforward of this morphism, see Lemma 0A9E. Consider the relative dualizing complex
$$\omega_X^\bullet = a(\mathcal{O}_{\text{Spec}(k)})$$
Compare with Remark 0B6S. Since f is proper we have $f^!(\mathcal{O}_{\text{Spec}(k)}) = a(\mathcal{O}_{\text{Spec}(k)})$ by definition, see Section 0A9Y. Applying Lemma 0AA3 we find that $\omega_X^\bullet$ is a dualizing complex. Moreover, we see that $\omega_X^\bullet$ and ω_X are as in Example 0AWI and as in Example 0AWJ.

Parts (1), (2), and (3) follow from Lemma 0AWN.

For a closed point $x \in X$ we see that $\omega_{X,x}^\bullet$ is a normalized dualizing complex over $\mathcal{O}_{X,x}$, see Lemma 0AWL. Part (4) then follows from Dualizing Complexes, Lemma 0AWR.

Part (5) holds by construction as a is the right adjoint to $Rf_* : D_{QCoh}(\mathcal{O}_X) \to D(\mathcal{O}_{\text{Spec}(k)}) = D(k)$ which we can identify with $K \mapsto R\Gamma(X, K)$. We also use that the derived category $D(k)$ of k-modules is the same as the category of graded k-vector spaces.

Part (6) follows from Lemma 0AWP for coherent $\mathcal{F}$ and in general by unwinding (5) for $K = \mathcal{F}[0]$ and $i = -\dim(X)$.

Part (7) follows from Lemma 0BRT. □

[1]This property characterizes $\omega_X^\bullet$ in $D_{QCoh}(\mathcal{O}_X)$ up to unique isomorphism by the Yoneda lemma. Since $\omega_X^\bullet$ is in $D^b_{Coh}(\mathcal{O}_X)$ in fact it suffices to consider $K \in D^b_{Coh}(\mathcal{O}_X)$.